and one if $z \in \Sigma_E \setminus \mathcal{A}$ (the symbol $\setminus$ means the removal of the second set from the first). Then for any trajectory starting in $\mathcal{A}$, we have that the average of χ along Γ vanishes, i.e.

$$\bar{\chi}_\Gamma = \lim_{\tau\to\infty} \tau^{-1} \int_0^\tau \chi(z(t))\mathrm{d}t = 0,$$

since $\mathcal{A}$ is an invariant set.

On the other hand, the average of χ on Σ_E is just the probability measure of $\Sigma_E \setminus \mathcal{A}$. If we assume the system is ergodic, we must have $\Pr(\Sigma_E \setminus \mathcal{A}) = 0$, thus $\Pr(\mathcal{A}) = 1$. Since, according to the definition, equality of time and space averages is allowed to fail only on a set of probability measure zero, we have that either $\Pr(\mathcal{A}) = 0$ or $\Pr(\mathcal{A}) = 1$.

If a system is not ergodic, then there will be an invariant set $\mathcal{A}$ whose probability measure is strictly between zero and one. The trajectories emanating from this set are unable to explore the entire energy surface and the corresponding trajectory averages will, at least for some observables, be different than the spatial average of that observable. A system which is not microcanonically ergodic has an energy surface that can be partitioned into subdomains of nonzero measure in which trajectories are forever trapped.

Often, the strict definition of ergodicity must be relaxed to accommodate the realistic modelling setting; for example only some portion of the energy surface may be accessible by trajectories in the time-interval available, due to the cost of computing steps with a numerical method. For a more detailed discussion of these and other ergodicity issues in the context of deterministic molecular dynamics, see the articles of Tupper [379, 381].

Given a system with energy of the form $H(\boldsymbol{q},\boldsymbol{p}) = \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2 + U(\boldsymbol{q})$, what can be said about the ergodicity of the system on the energy surface $H = \mathrm{E}$? Coverage of the set may follow in certain cases from periodic or quasiperiodic motion in which the trajectories wrap around the surface filling in space.

As a very simple example, the harmonic oscillator is always ergodic on its energy surfaces (i.e. energy curves, in this case). Some examples of two-dimensional systems are shown in Fig. 5.3.

Ergodicity is a concept that often becomes more relevant in systems with many degrees of freedom. For example if one thinks of having not just one but many double well oscillators connected together in some way so that they can exchange energy, then it is likely that the energy can accumulate, given enough time, in one of the oscillators, enabling it to overcome the barrier.

Example 5.4 (2D Harmonic System) Consider a two-dimensional harmonic system, i.e. the system with Hamiltonian

$$H = p_1^2/2 + p_2^2/2 + \Omega_1^2 q_1^2/2 + \Omega_2^2 q_2^2/2,$$